

Research

Deep learning-based potato leaf disease classification and severity assessment

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Abstract

A deep learning model was developed to detect, classify, and assess the severity of potato leaf diseases using 4200 images captured with a smartphone. The dataset, including healthy and diseased leaves affected by early and late blight, was preprocessed with techniques like median filtering and data augmentation. For disease identification, three classes, early blight, healthy, and late blight, were targeted. For severity level detection, potato leaf images with early and late blight were categorized into six classes: EB-low, EB-moderate, EB-severe, LB-low, LB-moderate, and LB-severe. The proposed model's performance was compared with AlexNet and VGG16. The model, using a CNN architecture with 1 fully connected and 11 convolutional layers, achieved an overall accuracy of 99% for disease classification. For severity detection, the model reached a 96% average accuracy. Performance was compared to AlexNet and VGG16, with the proposed model outperforming both in all metrics. The softmax activation was used in the output layer to classify potato leaf diseases as well as categorize the severity level of diseased leaf images.

Keywords Deep learning · Potato disease classification · Severity level detection

1 Introduction

Potatoes are Ethiopia's leading root and tuber crop, covering 60% of the total area allocated to RTCs (296,578 hectares) and contributing 51% of total production (3.67 million tonnes) with an average yield of 13.9 metric tons per hectare (CSA, 2016; 2017/18). Despite their importance, potato yields remain low compared to neighboring countries like Rwanda and Kenya due to challenges such as late blight and early blight. Late blight (Fig. 1), caused by *Phytophthora infestans*, is particularly destructive, causing yield losses of 6.5–67.7% depending on conditions [1]. Its symptoms include dark lesions on leaves and stems, which can destroy a field in days under humid conditions. The similarity between late and early blight symptoms makes diagnosis difficult for farmers, often resulting in ineffective management [2, 3]. Current practices rely on labor-intensive visual assessments or laboratory tools, which are impractical for widespread use. Advanced and scalable solutions, such as precision agriculture and machine learning technologies, are essential to accurately diagnose and quantify disease severity, thereby improving potato productivity and supporting the livelihoods of smallholder farmers in Ethiopia's highlands.

Early blight, caused by *Alternaria solani* (and *Alternaria alternata*), affects stressed or aging potato plants. It starts with small, dark brown lesions on older leaves, often showing concentric rings. Infected leaves turn yellow and drop, causing

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Fig. 1 Late blight-affected potato leaf samples. All images represent samples of late blight-affected potato leaves



significant yield loss if uncontrolled. The disease severity varies with weather conditions, crop age, and potato maturity. Figure 2 shows early blight lesions on potato leaves [4, 5].

The severity of blight diseases is important for determining the amount of pesticide needed to control pests. Since blight affects the leaves, we classify the diseases based on their severity. According to [6] Late blight is categorized into four levels of severity. Table 1 shows the different severity levels of potato leaf late blight.

- **Healthy (0%):** Indicates that the potato plants show no signs of disease, which is the ideal state for healthy growth.
- **Mild (< 10%):** The presence of small patches may not severely impact the plant, but should be monitored to prevent progression.

Fig. 2 Early blight-affected potato leaf

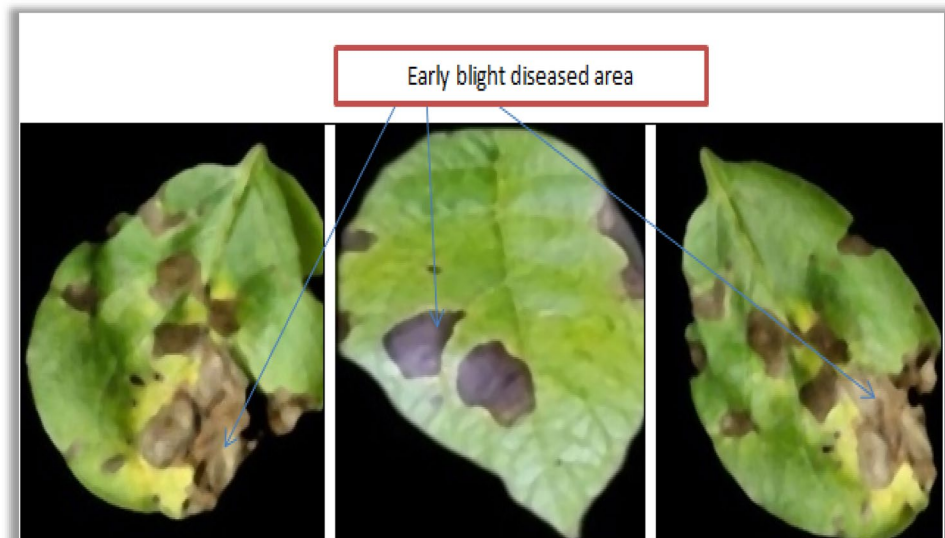


Table 1 Severity level of potato late blight leaf disease

Severity level	Category	Ratio (%)	Symptoms
0	Health	0	zero signs of disease
1	Mild	< 10	Tiny, erratic-shaped patches of light to dark green color
2	Moderate	10–50	On the leaf surface, there are large, dark brown blemishes
3	Severe	> 50	When the environment is moist, white mold grows on the leaf surface; when the environment is dry, the leaf often dries out and contracts

- Moderate (10–50%): This level signifies a concerning presence of disease, necessitating immediate intervention to avoid further damage.
- Severe (> 50%): At this stage, the disease significantly affects the plant's health, with visible mold and leaf damage. Immediate action is critical to prevent crop loss.

Figure 3 shows potato leaf images with different severity levels: (a) represents level 0 (health), (b) represents level 1 (mild), (c) represents level 2 (moderate), and (d) represents level 3 (severe disease of blight). In our situation, level 0 or health leaf is not included when estimating severity because we include it in disease type identification.

Accurate disease severity measurement is crucial for effective crop protection, as improper pesticide application can harm plants. Farmers often apply pesticides without considering the severity of the disease, leading to either overuse or insufficient treatment. To address this, we developed a deep learning model that classifies potato diseases and assesses their severity. The goal is to create an automated system that accurately identifies blight-affected leaves and determines disease severity, helping optimize pesticide use. In Ethiopia, where potatoes are a key crop, visual inspection is currently used, but it is costly, time-consuming, and prone to errors. Our approach aims to provide a more efficient and accurate solution.

2 Related works

Significant advancements have been made in automating plant disease identification using machine learning and deep learning techniques. Each study focuses on specific plants, datasets, or aspects of disease detection. This section reviews recent efforts in potato leaf disease classification and severity estimation.

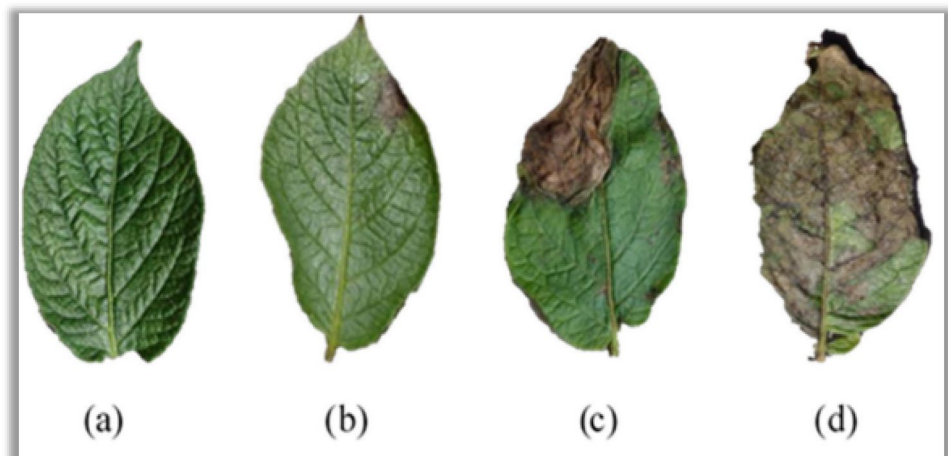
The author of [7] combined image segmentation and machine learning to diagnose potato diseases using 300 images from the PlantVillage dataset. They utilized a multiclass Support Vector Machine (SVM) classifier trained on color and texture features, achieving 95% accuracy in classifying potato leaves into healthy, early blight, and late blight categories. However, the study lacked real-environment datasets and severity analysis.

The author of [8] proposed a Backpropagation Neural Network (BPNN) model for classifying healthy and diseased potato leaves. Images were preprocessed with Gaussian filters, segmented using K-means clustering, and analyzed for features like color and texture using Gray Level Co-occurrence Matrix (GLCM). The model achieved over 92% accuracy but focused only on early blight and did not consider disease severity.

The author of [9] explored specialized CNN architectures for detecting diseases in multiple plants. Although their models effectively classified diseases, the approach did not estimate disease severity, a crucial metric for pesticide recommendations.

The author of [10] applied CNN architectures, including Inception v3, to classify carrot diseases using a dataset of 10,655 augmented images. Inception v3 achieved 97.4% accuracy. While effective for disease detection, the study omitted severity estimation, limiting its practical applicability.

Fig. 3 Severity level of potato leaf late blight disease



The author of [11] developed a CNN model for potato leaf diseases using PlantVillage data. The model achieved high classification accuracy but suffered from dataset limitations, including class imbalance and lack of real-world data, leading to higher false-positive rates in diverse environments.

The author of [12] applied fuzzy C-means clustering for late blight severity measurement in potato leaves. Although the model achieved 93% accuracy, it relied on manual disease identification and a small dataset, limiting its scalability.

The author of [13] proposed a 14-layer deep CNN architecture for potato disease classification, achieving 98% accuracy after augmenting a PlantVillage dataset from 1,722 to 9,822 images. However, the study did not address real-world environmental variability or severity estimation.

The author of [14] Introduced Cascaded Mask R-CNN for segmenting diseased regions and estimating disease severity in real-world images. The model achieved 91% Dice accuracy but required extensive manual labeling, limiting efficiency and scalability.

The author of [15] compared multitask CNNs and optimization strategies for coffee leaf disease classification and severity estimation. ResNet34 achieved high performance, but imbalanced datasets impacted the severity accuracy. They emphasized the need for balanced datasets for better generalization. We summarized the related works' methods, datasets, accuracy, and weaknesses in Table 2.

These studies demonstrate progress in plant disease classification but highlight gaps in real-world applicability, severity estimation, and handling dataset variability, forming the basis for further research in these areas.

The contribution of our study is:

The proposed model achieved 99% accuracy in disease classification, significantly outperforming AlexNet (96%) and VGG16 (95%). In severity estimation, it excelled with 96% accuracy, surpassing AlexNet (91%) and VGG16 (89%), showcasing its superior performance.

Key to this improvement were several unique architectural choices and training strategies. The model utilized a combination of 3×3 and 5×5 convolutional filters, which optimized feature extraction while reducing complexity. This architectural design allowed for more effective capture of relevant image features, particularly in distinguishing subtle differences in disease progression.

Additionally, we employed advanced data augmentation techniques during training, which enhanced the model's ability to generalize across varying conditions. Through using a diverse dataset collected from different environments in Ethiopia, the model demonstrated high precision and recall across all severity levels. It excelled in handling subtle disease progression for early and late blight, effectively differentiating between severe and low severity levels, an area where other models faced challenges.

With its high accuracy and robustness, the model offers significant practical benefits for precision agriculture. It facilitates targeted interventions and efficient resource management, with strong potential for deployment in mobile applications. The combination of architectural innovation and effective training strategies underscores the model's novelty and contribution to the field.

3 Research methodology

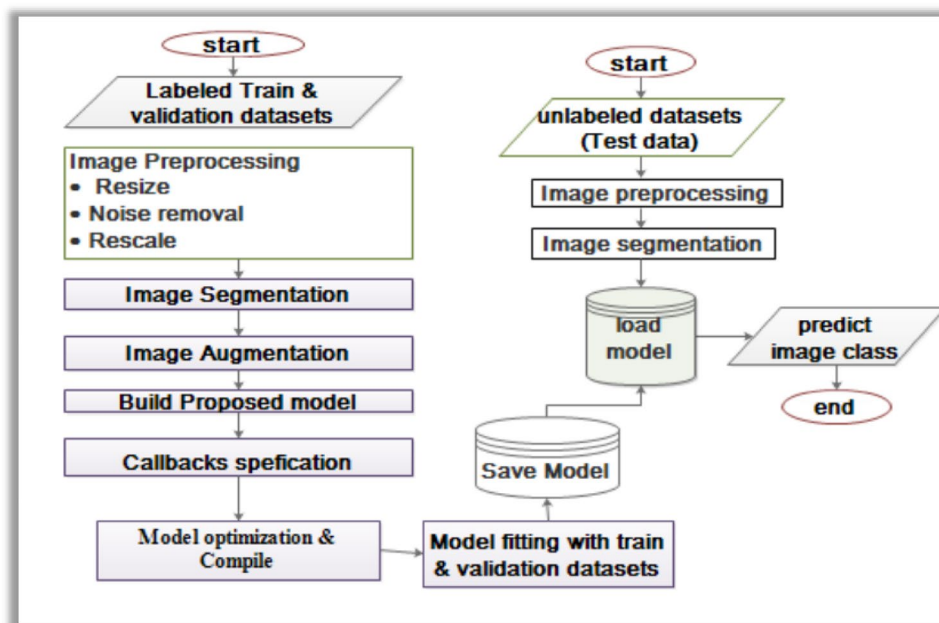
This study employs the Design Science Research (DSR) methodology, commonly used in Information Systems to develop innovative solutions for complex problems. DSR focuses on creating and evaluating artifact tangible systems or objects designed to address specific needs. It integrates the design process and fosters new research ideas and hypotheses [16].

3.1 Proposed system architecture

The proposed system architecture is divided into two components: (1) disease type identification and (2) severity level estimation. This separation ensures balanced data and prevents overfitting, as classifying both tasks simultaneously could lead to data imbalance. Figure 4 illustrates the process where input data undergoes preprocessing (filtering, noise removal, rescaling), followed by segmentation and augmentation. The model is developed using the Adam optimizer, compiled, and trained with the training and validation data. After training, the model is saved. Test data is preprocessed to match model parameters, and the saved model is used to predict the class of unlabeled images.

Table 2 Summary of related works

Author	Method	Dataset	Accuracy	Weakness
[7]	SVM with image segmentation and GLCM-based features	PlantVillage (300 images)	95%	Limited diseases (early/late blight), no severity analysis, lacks real-environment data
[8]	BPNN, Gaussian filter, K-means clustering	Custom dataset (healthy/diseased leaves)	> 92%	Focused only on early blight, no severity estimation
[9]	CNN-based deep learning models	Mixed sources (on-site and external)	Not specified	Did not estimate severity, lacked plant-specific disease insights
[10]	Inception v3 CNN on augmented carrot dataset	10,655 images (original + augmented)	97.4%	Excluded severity analysis, focused on carrots only
[11]	CNN on PlantVillage dataset	PlantVillage (1,000 + images/class)	Not specified	Dataset imbalance, no real-world dataset, and high false-positive rates in diverse environments
[12]	Fuzzy C-means clustering for severity measurement	Custom dataset (340 images)	93%	Manual disease identification, small dataset
[13]	14-layer CNN with data augmentation	PlantVillage (1,722 augmented to 9,822)	98%	No real-world dataset, no severity estimation
[14]	Cascaded Mask R-CNN for segmentation and severity	Real-world images (manually labeled)	Dice: 91%	Extensive manual labeling, poor scalability, and low precision
[15]	Multitask CNNs with ResNet34 and optimizers	Custom dataset (coffee leaves)	Not specified	The imbalanced dataset impacted severity estimation accuracy

Fig. 4 The proposed architecture

3.2 Data acquisition

The data was sourced from The Rare Farm at Haramaya University, an agricultural research area, serves as a potential hub for potato cultivation. Images of potato leaves were captured using a high-resolution smartphone camera (48 MP + 5 MP with f/2.0 and f/2.2 apertures). Plant pathology experts assisted in tagging the images for early blight, late blight, and healthy potato leaves. Additionally, they classified the images based on severity: "low", "moderate", and "severe" for both early and late blight. To address class imbalance and increase data diversity, image augmentation techniques expanded the dataset from 4,200 to 6,300 images.

3.3 Dataset preparation

To implement neural networks, the dataset must be split into training, validation, and testing subsets. The training dataset, comprising 70% (4,410 images), is used to train the model. The validation set (15% or 945 images) is used for model tuning, while the remaining 15% (945 images) is reserved for testing. Following guidelines [17], the training set comprises 60–90% of the total dataset, with the rest used for validation and testing. The potato leaf dataset consists of three categories: early blight, late blight, and healthy leaves. After applying data augmentation, the dataset was balanced and split as shown in Table 3.

Additionally, the dataset was expanded with specific severity classifications (low, moderate, and severe) for both early and late blight, totaling 5,214 images after augmentation. The dataset for severity estimation is presented in Table 4.

Table 3 Dataset for potato leaf disease identification

Class	Training (70%)	Validation (15%)	Testing (15%)
Early blight	1,470	315	315
Late blight	1,470	315	315
Healthy leaf	1,470	315	315
Total	6,300	945	945

Table 4 Potato leaf blight severity identification dataset

Class	Severity level	Training (70%)	Validation (15%)	Testing (15%)
Early blight (Low)	Low	541	163	165
Early blight (Moderate)	Moderate	541	163	165
Early blight (Severe)	Severe	541	163	165
Late blight (Low)	Low	541	163	165
Late blight (Moderate)	Moderate	541	163	165
Late blight (Severe)	Severe	541	163	165
Total		3,246	978	990

3.4 Image preprocessing

To make the dataset suitable for deep learning, the images undergo several preprocessing steps, including cropping, renaming, resizing, denoising, and segmentation. These steps ensure that the region of interest (ROI) is well-defined and the images are normalized for faster training.

Image Resize: The image dataset was initially large ($3000 \times 4000 \times 3$ pixels), making it computationally expensive. Therefore, all images were resized to 127×127 pixels to ensure compatibility with the proposed model and AlexNet (227×227 for AlexNet, 127×127 for our model), reducing computational burden.[18].

Image Denoise: The dataset was denoised using a median filter, which effectively removes "salt and pepper" noise from images. This improves image clarity and ensures better results in subsequent processing tasks [19].

Rescaling (Normalization): Image pixel values were normalized by dividing by 255, scaling the values between 0 and 1. This standardizes the images, making them easier for the neural network to process and minimizing issues with computational efficiency [20].

Data Labeling: Images were labeled according to their disease type (early blight, late blight, healthy) and severity level (low, moderate, severe). The labels for disease classification and severity identification are shown in Tables 5 and 6, respectively.

One-Hot Encoding: One-hot encoding was applied to convert categorical labels into binary vectors, ensuring the model can handle multi-class classification effectively. This encoding was applied to both disease classification and severity identification, as shown in Tables 7 and 8.

Data Augmentation: To prevent overfitting and increase dataset size, data augmentation techniques were applied using the Keras libraries. Transformations such as shear, zoom, width and height shift, brightness adjustment, and flipping were applied to augment the dataset during training, improving model robustness.

Table 5 Numerical labeling for disease identification

Numerical value	Class name
0	Early blight of potato leaves
1	Healthy potato leaves
2	Late blight of potato leaves

Table 6 Labeling for disease severity

Numerical value	Class name
0	EB-Low (Early Blight—Low)
1	EB-Moderate (Early Blight—Moderate)
2	EB-Severe (Early Blight—Severe)
3	LB-Low (Late Blight—Low)
4	LB-Moderate (Late Blight—Moderate)
5	LB-Severe (Late Blight—Severe)

Table 7 One-hot encoding for potato leaf disease classification

Early blight	Healthy leaf	Late blight
1	0	0
0	1	0
0	0	1

Table 8 One-hot encoding for disease severity identification

EB-low	EB-moderate	EB-severe	LB-low	LB-moderate	LB-severe
1	0	0	0	0	0
0	1	0	0	0	0
0	0	1	0	0	0
0	0	0	1	0	0
0	0	0	0	1	0
0	0	0	0	0	1

3.5 The proposed model

The proposed deep CNN model for potato disease classification and severity estimation includes a softmax layer, 1 dense layer, convolutional layers, and an image input layer. It employs convolutional layers with 3×3 and 5×5 filters, ReLU activation, batch normalization, max pooling, and dropout regularization after the third convolution and fourth pooling layers. The model consists of 13 layers in total. Four max pooling layers (2×2 filters) reduce computational costs, while fully connected layers (512 and 3, 6 units) classify disease and severity. The model is trained on 70% of the dataset, with 30% reserved for testing.

Input Layer: The input layer receives raw data and passes it through the network for processing. During training, it feeds data into the network and updates its weights to minimize prediction errors. It plays a crucial role in feeding the model with training data. Once processed, the data moves to subsequent layers for further analysis. The input layer is adjusted to improve model accuracy during training [21].

Convolutional Layers: Convolutional layers in CNNs extract features from the input data using learnable filters (kernels). These filters are convolved over the input to produce feature maps that capture local patterns. The convolution operation computes the dot product between the filter and local regions of the input. Hyperparameters like filter size and stride influence the output size. Convolution layers help the model recognize important features in images or signals [22].

ReLU Layer: ReLU (Rectified Linear Unit) introduces non-linearity by setting all negative input values to zero and retaining positive ones. This allows the neural network to model complex, non-linear relationships. It is commonly used in deep learning networks for its simplicity and efficiency. ReLU accelerates the learning process by promoting sparsity in the activations. It helps networks learn more complex patterns from the data [23].

MaxPooling Layer: MaxPooling is a down-sampling operation used to reduce feature map dimensions while retaining essential information. A pooling window slides over the input, selecting the maximum value in each region. This reduces computational load and prevents overfitting by abstracting important features. Typically, pooling sizes of 2×2 or 3×3 are used. It helps the model focus on the most significant features [24].

Dropout Layer: Dropout is a regularization technique that randomly "drops" neurons during training to prevent overfitting. It disables certain neurons by setting their outputs to zero, forcing the model to learn more robust features. This technique promotes generalization and reduces reliance on any specific neuron. Dropout is applied during training and turned off during testing. It improves the network's ability to generalize to unseen data [25].

Fully Connected Layer: A Fully Connected (Dense) Layer connects every neuron from the previous layer to every neuron in the current layer. It performs a weighted sum of inputs and applies an activation function to introduce non-linearity [26]. This layer helps the model combine learned features for complex decision-making. It allows the network to capture higher-level representations. Fully connected layers are often used in the final stages of a neural network.

Softmax Classifier: The Softmax function converts raw scores from the final layer into a probability distribution over classes. It normalizes the input values to produce probabilities between 0 and 1, where the sum equals 1. The

output represents the likelihood of each class, facilitating classification. Softmax is typically used in the output layer of classification models [27]. It enables the network to make a clear decision about which class the input belongs to.

Feature Extraction: Feature extraction involves transforming raw data into meaningful features for classification. It reduces data complexity by highlighting relevant information while discarding irrelevant details. In image classification, features like edges or textures are extracted. This step improves model efficiency and accuracy. Effective feature extraction helps the model focus on the most informative aspects of the data.

Training Phase: In the training phase, the CNN model learns to recognize patterns from the labeled training dataset. The training involves multiple epochs, where the model updates its weights and biases to minimize the loss function.

Validation Phase: In the validation phase, the CNN model is evaluated on a separate labeled dataset that is not used during training. This allows us to monitor the model's performance and detect overfitting. The validation dataset is used to tune the hyperparameters of the CNN, such as the learning rate, number of epochs, and batch size.

Testing Phase: In the testing phase, the model's performance is evaluated on unseen data to estimate its accuracy and effectiveness.

Table 9 outlines the basic layers of proposed model. The output layer consists of 3 and 6 units, representing the disease classes and disease severity levels, respectively. Additionally, Table 10 lists the specific hyperparameters applied to enhance the performance of the proposed model.

Evaluation Metrics: The effectiveness of machine learning models is evaluated using various metrics, which depend on the specific task and problem being addressed. To assess the performance of our model, we used the following well-known evaluation metrics:

Table 9 Layers of the proposed model

Layer	Type	No. filters	Filter size	Output	No. params
L1	Convolution	32	3 × 3	125 × 125 × 32	896
	Activation	Relu			
	MaxPooling				
L2	Convolution	32	5 × 5	58 × 58 × 32	25,632
	Activation	Relu			
	MaxPooling				
L3	Convolution	64	3 × 3	27 × 27 × 64	18,496
	Activation	Relu			
	Dropout (0.5)				
	MaxPooling				
L4	Convolution	128	5 × 5	9 × 9 × 128	204,928
	Activation	Relu			
	MaxPooling				
L5	Convolution			2048	
L6	Convolution	512	3 × 3	512	1,049,088
	Dropout (0.5)				
L7	Convolution	256		256	524,544
	Activation	Relu			
L8	Convolution	128		128	32,896
	Activation	Relu			
L9	Convolution	64		64	8,256
	Activation	Relu			
L10	Dropout (0.5)			64	
L11	Convolution	32		32	2,080
	Activation	Relu			
L12	Convolution	16		16	528
	Activation	Relu			
L13	Dense	8		8	136
	Activation	Relu			
L14	Softmax	3, 6		6	54

Table 10 Hyperparameters of the proposed model

Parameters	Values
Batch size	16
Epochs	100
Learning rate	0.0001–0.0000001
Loss function	Categorical cross-entropy
Dropout layer	0.5
Optimizers	Adam

Confusion Matrix: This matrix displays the actual versus predicted class labels, with entries showing the number of data points that fall into each combination of true and predicted classifications. True Positives (TP): Correctly classified positive instances. False Positives (FP): Incorrectly classified as positive. True Negatives (TN): Correctly classified negative instances. False Negatives (FN): Incorrectly classified as negative.

We specified that the precision (P), recall (R), and F1 scores reported in our results are macro-averaged. This approach was chosen to provide a balanced view of the model's performance across all severity levels, ensuring that each class contributes equally to the overall performance metrics.

We included a detailed explanation of the Area Under the Curve (AUC) interpolation methods used for calculating the mean Average Precision (mAP). We clarified that we employed the trapezoidal rule for interpolation, which ensures a smooth estimation of the precision-recall curve across various thresholds.

We added a section discussing the mAP50-95 metric, which assesses the model's performance across multiple Intersection over Union (IoU) thresholds from 0.50 to 0.95. This metric provides a more comprehensive evaluation of the model's ability to detect and classify objects accurately at varying levels of overlap. We included comparative results for mAP50-95 alongside our existing metrics, reinforcing the robustness of our model.

Table 11 summarizes the key evaluation metrics used to assess the performance of the model in terms of classification accuracy, precision, recall, F1-score, and macro averages [28].

Proposed Model Evaluation: To assess the performance of the proposed model, we conducted comparisons with well-established deep learning models such as AlexNet and VGG16. The evaluation was based on key criteria as presented in Table 12.

Table 11 Evaluation metrics

Metric	Formula	Description
Accuracy	$A = \frac{(TP+TN)}{TP+TN+FP+FN}$	Proportion of correct classifications
Precision	$P = \frac{TP}{TP+FP}$	Proportion of true positives among all positive predictions
Recall (Sensitivity)	$\frac{TP}{TP+FN}$	Proportion of true positives among all actual positives
F1-score	$1 - \text{score} = \frac{2 \times (\text{precision} \times \text{recall})}{(\text{precision} + \text{recall})}$	Harmonic mean of precision and recall
Precision macro-average	Precision macro – average = $\frac{(p_1 + p_2 + p_3 + \dots + p_N)}{N}$	Average precision across all classes
Recall macro-average	Recall macro – average = $\frac{(R_1 + R_2 + R_3 + \dots + R_N)}{N}$	Average recall across all classes

Table 12 The evaluation criteria of the proposed model

Criterion	Description
Accuracy	A higher accuracy model is preferred for detecting and classifying potato blight leaf diseases and assessing disease severity
Model loss	A lower loss value indicates a model's ability to generalize well, especially for new images not included in the training and validation sets
Model size	A model with smaller storage requirements is more efficient, particularly in resource-constrained environments
Model parameters	A model with fewer trainable parameters is easier and faster to train, validate, and test

3.6 Results and discussion

This section presents the evaluation results of the proposed model for the automatic classification of potato blight leaf diseases and the determination of disease severity levels.

3.7 Description of the dataset used

The dataset was prepared, segmented, and transformed before being divided into training, validation, and testing sets. Random splits were used to allocate 70% of the data for training, with the remaining 30% split equally into testing and validation sets (15% each).

3.8 Training methods

The training was conducted in two phases: Phase 1: Disease Diagnosis: The model classified potato leaves as early blight, late blight, or healthy using both the proposed model and transfer learning (AlexNet and VGG16). Phase 2: Severity Classification: The model classified the severity levels of early and late blight (early low, early moderate, early severity, late low, late moderate, and late severity).

3.9 Training pretrained models and the proposed model

Using transfer learning, we froze some layers of pretrained models and retrained the remaining layers for our specific task.

Experiment 1: Training VGG16 Model: The performance was analyzed for the VGG16 model, which was trained using only the last layer of the pre-trained model (transfer learning). The results of training VGG16 summarized below. The similar values for training and validation accuracy suggest that the model is not overfitting and the model generalizes well to unseen data. The classification report of the model is presented in Table 13.

Analysis: The model correctly identified 95% of early blight images, with 5% misclassified as either healthy or late blight. 96% of healthy images were correctly classified, with 4% misclassified as early blight or late blight. 96% of late blight images were accurately classified, with 4% misclassified as either early blight or healthy. Overall, the classifier achieved 95% accuracy using the dataset, showing robust performance in classifying all three categories: early blight, healthy leaves, and late blight. The model performs particularly well in classifying healthy leaves and early blight, with slightly less precision for late blight. Figure 5 illustrates the confusion matrix report of the VGG16 model in disease classification.

Experiment 2: Training AlexNet model: The model's performance metrics and classification results were evaluated to determine its effectiveness in identifying early blight, healthy leaves, and late blight diseases in potato leaves. The similar values between training accuracy and validation accuracy suggest that the model is not overfitting. The model performed well during training and validation, achieving a strong test accuracy of 95.92%, which indicates its generalizability across unseen data. The classification report of the model is presented in Table 14.

Analysis: The model correctly identified 97% of early blight images, with only 3% misclassified as healthy or late blight. 100% of healthy images were correctly identified, with no misclassifications. 91% of late blight images were correctly identified, while 9% were misclassified as early blight or healthy. The classifier achieved an accuracy of 96% overall, which is a strong performance, especially for early blight and healthy classifications. However, the model

Table 13 Classification report using VGG16

Class	Precision	Recall	F1-score	Support
Early blight	0.96	0.95	0.96	315
Healthy	0.99	0.96	0.97	315
Late blight	0.91	0.96	0.94	315
Accuracy	–	–	0.95	945
Macro average	0.95	0.95	0.96	945
Weighted avg	0.95	0.95	0.96	945

Fig. 5 Confusion matrix report of VGG16 trained

		Predicted		
		Early Blight	Healthy	Late Blight
Actual	Early Blight	299	0	16
	Healthy	0	302	13
	Late Blight	11	3	311

Table 14 Classification report of AlexNet model

Class	Precision	Recall	F1-score	Support
Early blight	0.97	0.97	0.97	315
Healthy	0.93	1.00	0.96	315
Late blight	0.98	0.91	0.94	315
Accuracy	–	–	0.96	945
Macro average	0.96	0.96	0.96	945
Weighted avg	0.96	0.96	0.96	945

Fig. 6 Confusion matrix report for fully trained AlexNet

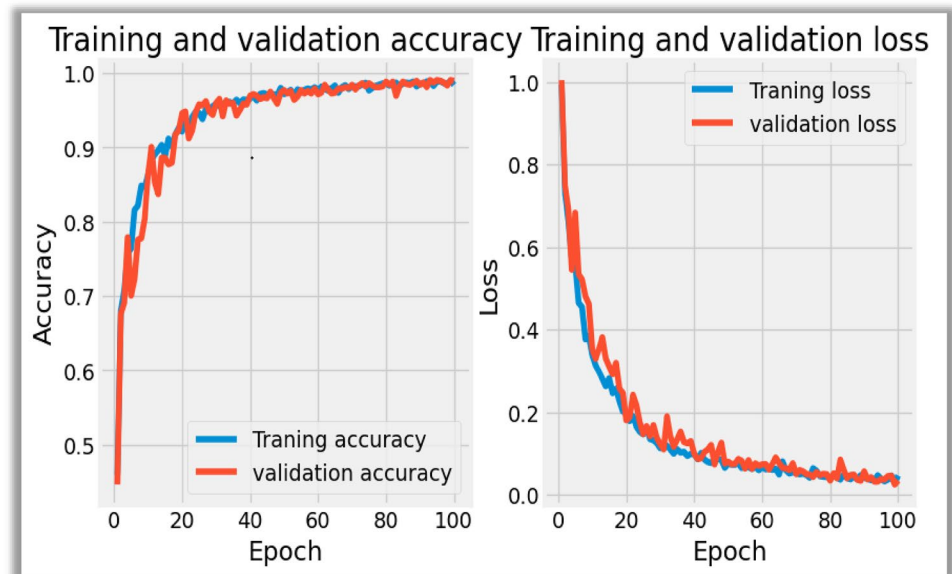
		Predicted		
		Early Blight	Healthy	Late Blight
Actual	Early Blight	316	4	5
	Healthy	0	315	0
	Late Blight	9	21	286

showed slightly lower performance in classifying late blight, where 9% of the images were misclassified as healthy or early blight. Figure 6 illustrates the confusion matrix report of the AlexNet model in disease classification.

Experiment 3: Training the Proposed Model: The model's performance was evaluated on early blight, healthy, and late blight potato leaf images, and the results indicate a high level of classification accuracy. Given the similar values between training accuracy and validation accuracy, we can confidently conclude that the model is not overfitting.

Table 15 Classification report of the proposed model

Class	Precision	Recall	F1-score	Support
Early blight	0.97	0.99	0.98	315
Healthy	1.00	1.00	1.00	315
Late blight	0.99	0.97	0.98	315
Accuracy	–	–	0.99	945
Macro average	0.99	0.99	0.99	945
Weighted avg	0.99	0.99	0.99	945

Fig. 7 The training and validation curve of the proposed model

This suggests that the proposed model is generalizing well to unseen data. The classification report of the proposed model is presented in Table 15. Figure 7 illustrates the training and validation curve of the proposed model.

Analysis: The proposed model was able to classify 99% of early blight cases correctly. The remaining 1% of images were misclassified as healthy or late blight. The model correctly identified 100% of healthy images, with no misclassifications. The model accurately classified 97% of late blight images. However, 3% of the images were incorrectly identified as early blight or healthy. The proposed model achieved an overall classification accuracy of 99%, which is exceptionally high. It performed particularly well in identifying healthy images with perfect accuracy (100%). For early blight, the accuracy was also very high at 99%, but a small portion of images was misclassified as healthy or late blight. Similarly, for late blight, while 97% of the images were accurately classified, 3% were misclassified as early blight or healthy. Despite the small errors, particularly in distinguishing between early blight and late blight, the model demonstrated strong performance overall, and the misclassifications are likely due to the similarities in the symptoms of early and late blight diseases. This suggests that the model is effective in most cases but could benefit from further refinement to better handle edge cases where the symptoms overlap. Figure 8 illustrates the confusion matrix report of the proposed model in potato disease classification.

3.10 Severity identification

After successfully classifying potato leaf diseases into early blight (EB), healthy, and late blight (LB) categories, the next step is severity estimation. This involves determining the level of disease severity for the affected images.

Experiment 4: Training AlexNet model to estimate severity: The severity levels were divided into three categories for each disease: low, moderate, and severe. The severity classification report of AlexNet is presented in Table 16

Analysis of Severity Estimation: Early Blight (EB) Severity: 99% of low-severity early blight cases were accurately estimated, with only 1% misclassified as other severity levels. The model correctly estimated 91% of moderate-severity early

Fig. 8 Confusion matrix report of the proposed model

		Predicted		
		Early Blight	Healthy	Late Blight
Actual	Early Blight	313	0	2
	Healthy	0	315	0
	Late Blight	9	1	305

Table 16 Severity classification report of AlexNet

Class	Precision	Recall	F1-score	Support
EB-low	0.92	0.99	0.95	165
EB-moderate	0.94	0.91	0.92	165
EB-severe	0.98	0.96	0.97	165
LB-low	0.81	0.99	0.89	165
LB-moderate	0.85	0.73	0.78	165
LB-severe	0.96	0.85	0.90	165
Accuracy	–	–	0.91	990
Macro average	0.91	0.91	0.90	990
Weighted average	0.91	0.91	0.90	990

blight images, with 9% incorrectly classified as another severity level. 96% of severe early blight cases were accurately classified, while 4% were misclassified as other severity levels. Late Blight (LB) Severity: 99% of low-severity late blight images were correctly classified, with 1% misclassified as other severity levels. 73% of moderate-severity late blight images were correctly classified, but 27% were misclassified as other severity levels. Severe (LB-severe): 85% of severe late blight images were accurately classified, with 15% misclassified. Figure 9 illustrates the confusion matrix report of the AlexNet model in potato disease severity classification.

Experiment 5 Training VGG16 for Severity Identification: To further evaluate severity estimation, we trained the VGG16 model to assess the impact of this fine-tuning on the severity estimation task. The severity classification report of VGG16 is presented in Table 17

Analysis of Severity Estimation with VGG16: Early Blight (EB) Severity: 95% of low-severity early blight cases were correctly identified, with 5% misclassified as other severity levels. 92% of moderate-severity early blight images were correctly classified, while 8% were misclassified. 98% of severe early blight images were correctly identified, with 2 images misclassified as other severity levels. Late Blight (LB) Severity: 77% of low-severity late blight images were correctly estimated, with 23% misclassified as other severity levels. 84% of moderate-severity late blight images were correctly classified, while 16% were misclassified as other severity levels. 86% of severe late blight images were correctly identified, with 14% misclassified. Figure 10 illustrates the confusion matrix report of the VGG16 model in potato disease severity classification.

Experiment 6 Proposed Model Severity Estimation: The proposed model accurately classified severity levels for both early blight (EB) and late blight (LB) diseases. These values indicate that the proposed model does not exhibit

Fig. 9 Confusion matrix report of AlexNet in severity classification

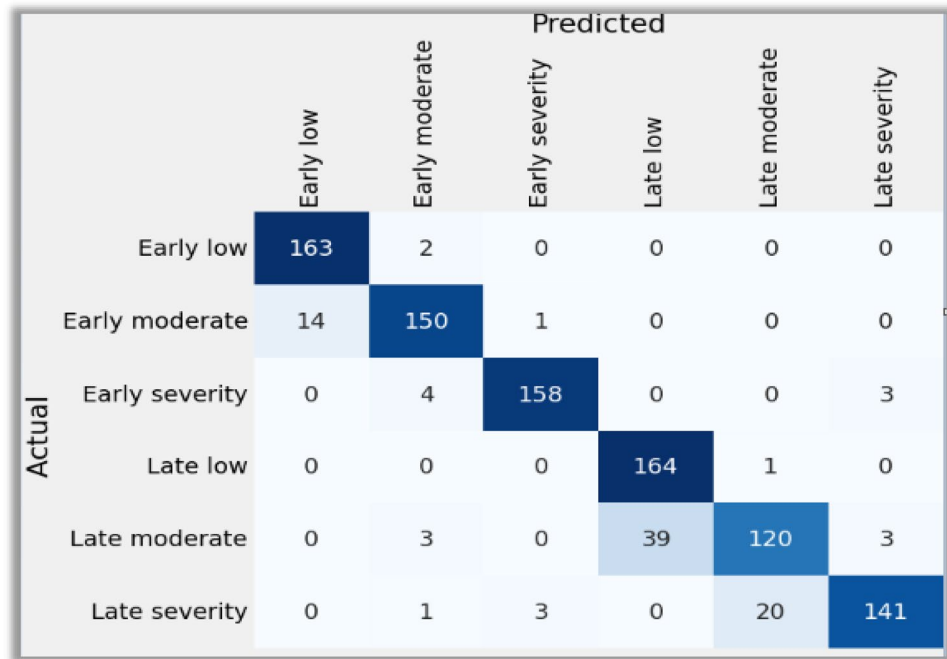


Table 17 Severity classification report of the VGG16 model

Class	Precision	Recall	F1-score	Support
EB-low	0.91	0.95	0.93	165
EB-moderate	0.89	0.92	0.91	165
EB-severe	0.98	0.98	0.98	165
LB-low	0.94	0.77	0.85	165
LB-moderate	0.73	0.84	0.78	165
LB-severe	0.92	0.86	0.89	165
Accuracy	-	-	0.89	990
Macro average	0.89	0.89	0.89	990
Weighted average	0.89	0.89	0.89	990

Fig. 10 Confusion matrix report of VGG16 in severity Identification

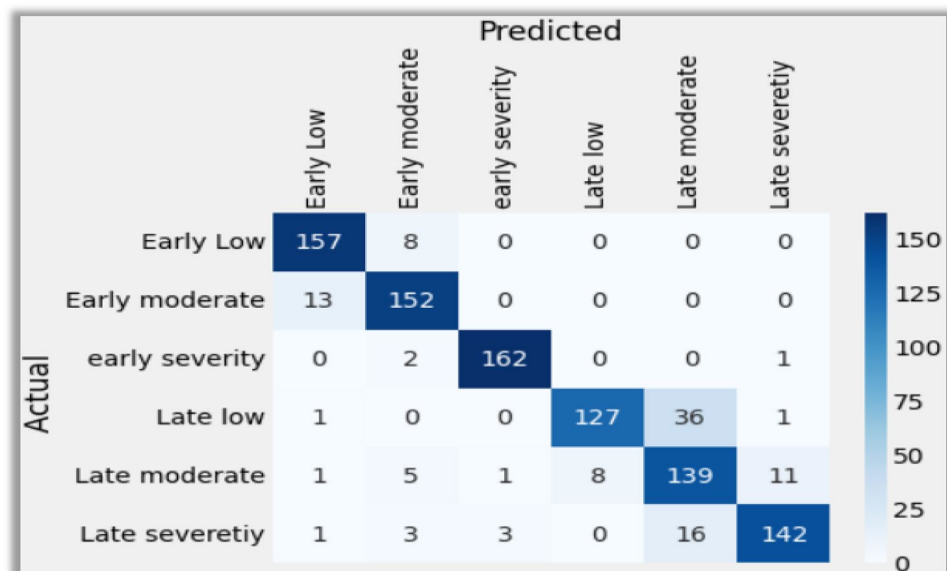


Table 18 Severity classification report of VGG16 model

Class	Precision	Recall	F1-score	Support
EB-low	1.00	0.98	0.99	165
EB-moderate	0.97	1.00	0.99	165
EB-severe	0.99	0.99	0.99	165
LB-low	0.94	0.98	0.96	165
LB-moderate	0.94	0.90	0.92	165
LB-severe	0.96	0.94	0.95	165
Accuracy	–	–	0.96	990
Macro average	0.96	0.96	0.96	990
Weighted average	0.96	0.96	0.96	990

Fig. 11 Confusion matrix report severity estimation of the proposed model

		Predicted					
		Early low	Early moderate	Early severity	Late low	Late moderate	Late severity
Actual	Early low	161	4	0	0	0	0
	Early moderate	0	165	0	0	0	0
	Early severity	0	0	164	0	0	1
	Late low	0	0	0	161	3	1
	Late moderate	0	0	0	11	149	5
	Late severity	0	1	2	0	7	155

significant overfitting, as the training and validation accuracies are very close, suggesting good generalization to unseen data. The severity classification report of the proposed model is presented in Table 18

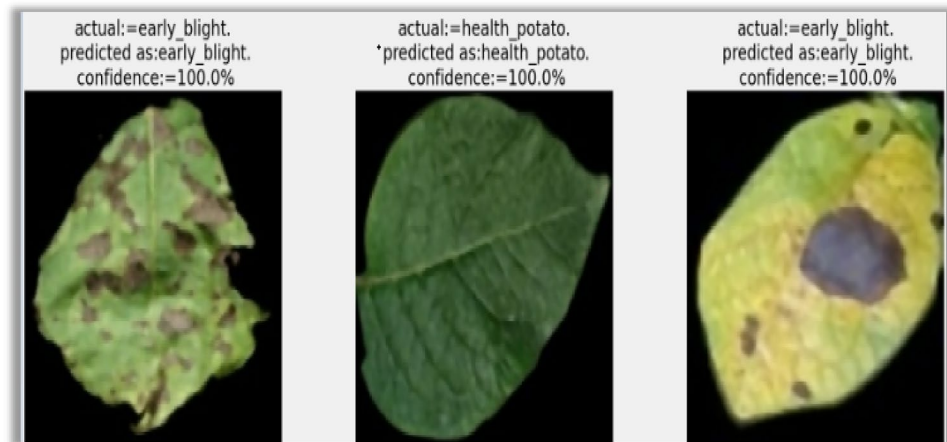
Analysis of Severity Estimation with the Proposed Model: Early Blight (EB) Severity: 98% of low-severity early blight cases were correctly classified, with 2% misclassified as other severity levels. The model perfectly classified 100% of moderate-severity early blight images. 99% of severe early blight cases were accurately identified, with 1% misclassified. Late Blight (LB) Severity: 98% of low-severity late blight cases were correctly classified, while 2% were incorrectly labeled. 90% of moderate-severity late blight cases were correctly classified, with 10% misclassified as other severity levels. 94% of severe late blight cases were correctly classified, with 6% misclassified. Figure 11 illustrate the confusion matrix report of the proposed model in potato disease severity classification.

Figure 12 illustrate some severity classification using the proposed model in potato leaf disease.

The following table provides a comprehensive comparison of the performance metrics across all models used for disease classification and severity estimation, highlighting their precision, recall, F1-scores, and accuracy rates for each specific task. The overall comparison of the study is presented in Table 19

Prototype Development: A user-friendly web interface using Gradio (Figs. 13 and 14) allows easy access for domain experts to classify potato leaf diseases and estimate their severity in real time. This tool could be useful in practical applications, where experts can quickly deploy the model without needing advanced technical skills.

Fig. 12 Severity classification in potato leaf disease



Prototype Development: We developed a user-friendly web interface using Gradio (see Fig. 13), which enables domain experts to classify potato leaf diseases and estimate their severity in real-time. This tool is designed for practical applications, allowing experts to deploy the model without requiring advanced technical skills.

Integration into Practical Farming Environments: The model's deployment in real-world agricultural settings involves several considerations to ensure its effectiveness:

1. **Adaptability to Lighting Conditions:** The model incorporates techniques to handle varying lighting conditions, which are common in outdoor environments. By employing data augmentation during training, including adjustments for brightness and contrast, the model learns to recognize symptoms under different lighting scenarios. This adaptability is crucial for accurate classification during different times of the day or in varying weather conditions.
2. **Handling Variable Image Quality:** Recognizing that image quality can vary due to factors such as distance, focus, and camera specifications, the model was trained on a diverse dataset that included images taken under various conditions. This approach enhances its robustness, enabling it to perform well even when images are less than ideal. Users can upload images captured with smartphones or other devices, and the model's performance remains reliable.
3. **Environmental Factors:** The model is designed to function in diverse agricultural environments, accounting for factors such as humidity, temperature, and background noise. By training on data collected from different locations in Ethiopia, the model learned to differentiate between disease symptoms and variations caused by environmental conditions. This ensures that the model can effectively classify diseases regardless of the specific agricultural context.
4. **User Engagement and Training:** To further facilitate the model's deployment, we propose training sessions for farmers and agricultural experts. These sessions would focus on how to use the web interface effectively and interpret the model's outputs. By guiding the tool's use, we enhance its practicality and ensure that users can make informed decisions based on the model's classifications.
5. **Continuous Feedback Loop:** We also envision incorporating a feedback mechanism where users can report back on the model's predictions. This would allow for continuous learning and improvement of the model, as real-world data can refine its accuracy and adaptability over time.

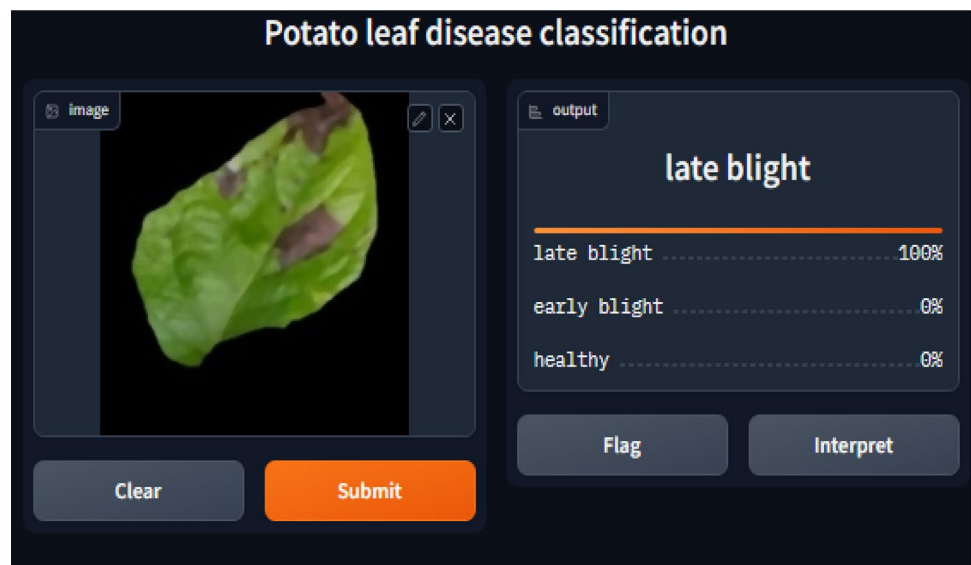
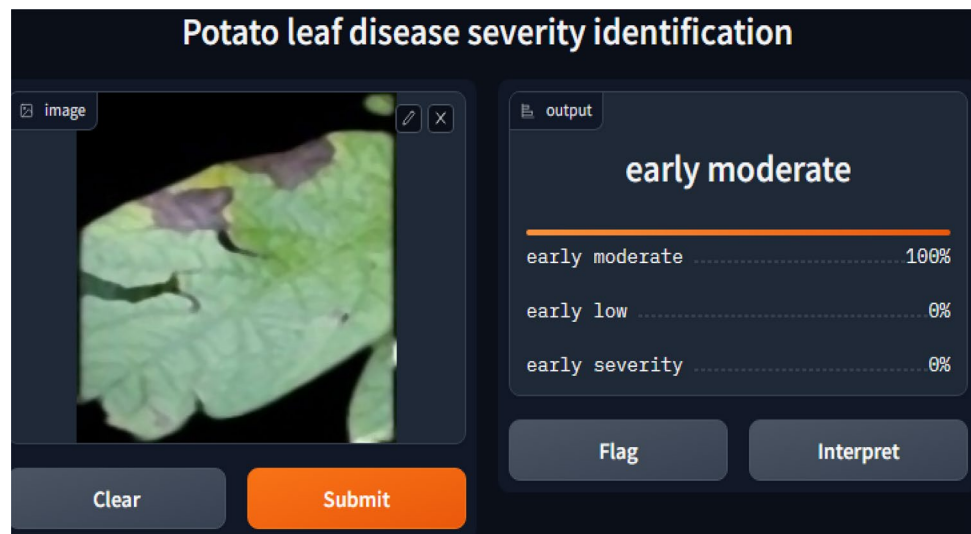
Table 20 demonstrates the classifier model's impressive performance, achieving 99% classification accuracy and 96% severity estimation accuracy. In conclusion, the proposed model outperforms previous studies in terms of robustness and accuracy in classifying potato leaf diseases and estimating severity. This superiority can be attributed to the reduced complexity of our model, which employs fewer parameters through the use of 3×3 and 5×5 filter sizes. These filter sizes effectively capture and extract relevant features from the images, enhancing the model's ability to differentiate between various disease states.

4 Discussion

The experiments conducted for disease classification and severity estimation of potato leaf diseases demonstrate the superior performance of the proposed model compared to AlexNet and VGG16-based models. The proposed model achieved an overall accuracy of 99% in disease classification, surpassing AlexNet's 96% and VGG16's 95%. It exhibited

Table 19 Overall comparison of models

Task	Model	Precision	Recall	F1-score	Accuracy (%)	Remarks
Disease classification	AlexNet	0.96	0.96	0.96	96	Strong generalization, but struggled slightly with late blight cases
	Proposed model	0.99	0.99	0.99	99	Best performer; consistently identified all disease types with high accuracy
	VGG16l	0.95	0.95	0.95	95	Slightly lower recall for some cases; good performance overall
Severity estimation (Early blight)	AlexNet	0.92–0.98	0.91–0.99	0.92–0.97	91	EB-severe had the highest F1-score; some confusion for EB-moderate
	VGG16	0.91–0.98	0.92–0.98	0.91–0.98	89	Lower accuracy in distinguishing LB-low and LB-moderate levels
	Proposed model	0.97–1.00	0.98–1.00	0.99	96	High performance, particularly for EB-severe and EB-moderate cases
Severity estimation (Late blight)	AlexNet	0.81–0.96	0.73–0.99	0.78–0.90	91	Performance dropped significantly for LB-moderate cases
	VGG16	0.73–0.92	0.77–0.86	0.78–0.89	89	Similar challenges in LB-moderate estimation accuracy
	Proposed model	0.90–0.96	0.90–0.98	0.92–0.96	96	Balanced and superior performance for LB-severe and LB-low cases

Fig. 13 Image displayed using the prototype**Fig. 14** Image displayed using the prototype for severity identification**Table 20** Summary of comparison of our model with similar work

Authors	Title	ML/DL tool	Classification accuracy (%)	
			Disease type	Severity level
Khalifa et al. [13]	Deep learning architecture for potato leaf blight classification	CNN	98	–
Athanikar & Badar, [29]	Potato leaf disease detection and classification system	ANN	92	–
Esgario et al. [30]	Classification and severity estimation of coffee leaf biotic stress	CNN	95.24	86.51
Lucero et al. [15]	Plant disease classification and severity estimation	Multitask CNN + 1st Order Optimizers	87.3	–
Biswas et al. [31]	Severity identification of potato late blight disease from Crop images captured under uncontrolled environment	ANN		93
Our model	CNN	99	96	

minimal misclassifications across all categories (early blight, late blight, and healthy), highlighting its robustness in distinguishing even visually similar disease symptoms. While AlexNet and VGG16I performed reasonably well, they showed some limitations, particularly in differentiating between early and late blight, where overlapping features posed challenges. In severity estimation, the proposed model again excelled with an accuracy of 96%, outperforming AlexNet (91%) and VGG16 (89%). For early blight severity levels, the proposed model showed high precision and recall across all categories (low, moderate, and severe), demonstrating its ability to handle subtle variations in disease progression. Similarly, for late blight severity levels, the model maintained consistent performance, particularly in distinguishing severe and low severity cases, which were challenging for other models. However, all models faced difficulties in accurately predicting moderate severity levels, especially for late blight, due to overlapping features and variability in the dataset.

5 Conclusion

The proposed approach introduced enhancements to the basic CNN architecture. The developed system demonstrated high precision and reliability, achieving 99.12% training accuracy, 98.94% validation accuracy, and 98.94% testing accuracy for disease identification. For severity estimation, the model achieved 95.93% training accuracy, 94.99% validation accuracy, 94.98% testing accuracy, and 96% average accuracy. However, the model occasionally misclassified disease types and severity levels due to the similarity of symptoms between early and late blight and the subtle differences in disease severity hues. These challenges highlight areas for future improvement to enhance the model's performance and reliability. Overall, the study established a robust framework for automated potato blight disease management, offering a practical and efficient solution for the agricultural sector.

6 Limitation

We acknowledge several challenges that impact the model's performance and overall applicability:

1. **Similarity Between Early and Late Blight Symptoms:** One of the primary challenges we encountered was the similarity in symptoms exhibited by early and late blight. This overlap can lead to misclassifications, particularly in cases where the symptoms are subtle. While the model achieved high accuracy, there remain instances where distinguishing between these two conditions proves difficult.
2. **Distinguishing Moderate Severity Levels:** The model also faced challenges in accurately classifying moderate severity levels. The nuanced differences in symptomatology at this stage can be hard to detect, which may affect the precision of severity estimation. We recognize that further refinement is needed to improve classification in this critical area.

Suggestions for Future Work:

To address these limitations and enhance the model's robustness, we propose the following avenues for future research:

1. **Incorporation of Additional Data:** Expanding the dataset to include images from a wider variety of growing conditions will be crucial. This should encompass different geographical regions, climate conditions, and growth stages of potatoes. A more diverse dataset will help the model learn to recognize symptoms under varying circumstances, thereby improving classification accuracy.
2. **Advanced Feature Extraction Techniques:** Implementing more sophisticated feature extraction methods, such as using attention mechanisms or ensembles of different models, could enhance the model's ability to discern subtle differences in symptoms. This approach may help in better distinguishing between similar disease states.
3. **User Feedback Integration:** Establishing a feedback loop where users can report discrepancies in the model's predictions will facilitate continuous learning. Incorporating real-world feedback can refine the model over time, leading to improved accuracy and reliability.
4. **Real-World Testing:** Conducting field trials to test the model under practical agricultural conditions will provide insights into its performance across various environments. This real-world validation will not only help identify limitations but also guide further enhancements.

5. Collaboration with Agricultural Experts: Engaging with agronomists and plant pathologists can provide deeper insights into symptomology and disease progression. Their expertise can inform future iterations of the model, ensuring that it aligns closely with real-world agricultural practices.

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Data Availability The datasets generated during and/or analyzed during the current study are not publicly available but are available from the corresponding author upon reasonable request.

Code availability Source code is available at: <https://github.com/Tadesse-Assefa/potato-leave-disease-research-code>.

Declarations

Ethics approval and consent to participate Not applicable.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

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